S16905

Individual Contribution: Model Evaluation using ROC Curve and AUC

- As part of the youth unemployment analysis in Sri Lanka, I introduced an additional model evaluation method using the ROC Curve and Area Under the Curve (AUC). This analysis was added to evaluate how well the logistic regression model distinguishes between employed and unemployed youth.

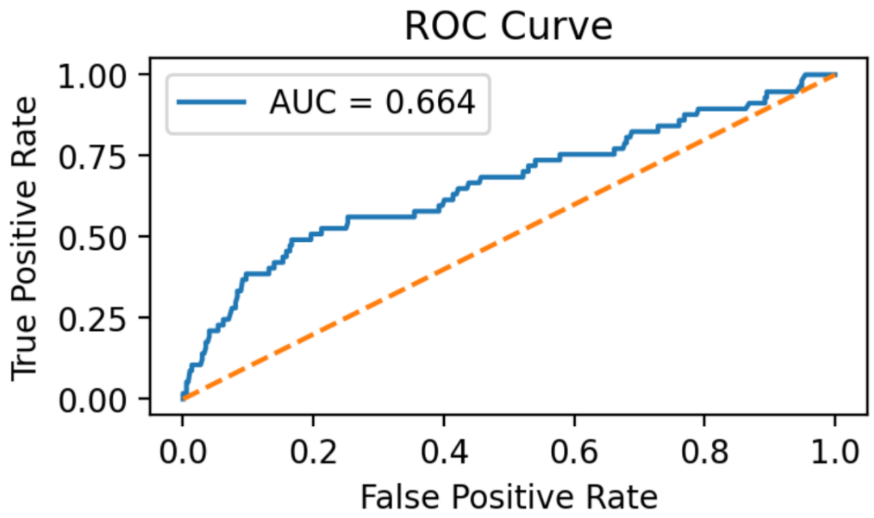
- The ROC Curve and AUC evaluation are used as an alternative method for evaluating the model's performance. The ROC stands for Receiver Operating Characteristic, and it plots the True Positive Rate against the False Positive Rate for various classification thresholds. The ROC Curve gives a graphical view of how well the model is classifying two different classes. The AUC value represents the overall performance of the model. The model has achieved an AUC value of 0.664. The range of the AUC value is between 0 and 1. The value 0.5 represents random guessing. The higher value represents better performance. The value 0.664 represents that the model is performing better than random guessing and has a moderate level of performance in distinguishing between unemployed and employed youth. The ROC Curve and AUC evaluation provide more reliability to the model because it is using more than one method for evaluating its performance.

📈 Model Evaluation – ROC Curve & AUC

This section adds a small new analysis to evaluate the logistic regression model. We use the ROC curve and compute the AUC to measure how well the model distinguishes between employed and unemployed youth.

AUC Score

0.664



Interpretation

The ROC curve shows how well the logistic regression model can distinguish between unemployed and employed youth.

- The **AUC (Area Under the Curve)** score ranges from 0 to 1:
  - 0.5 → The model is guessing randomly.
  - Closer to 1 → The model has better discrimination ability.
- Here, the AUC value indicates the model performs better than random guessing, meaning it has a moderate ability to separate unemployed from employed youth.

On the ROC graph:

- **True Positive Rate (TPR)** represents the percentage of unemployed youth correctly predicted as unemployed.
- **False Positive Rate (FPR)** represents the percentage of employed youth incorrectly predicted as unemployed.

In simple terms:

- A higher TPR is desirable because it means the model successfully identifies more unemployed youth.
- A lower FPR is desirable because it reduces incorrect classification of employed youth as unemployed.

Although the model is not perfect, the ROC analysis shows that it captures meaningful patterns in the data and provides additional evaluation beyond accuracy.